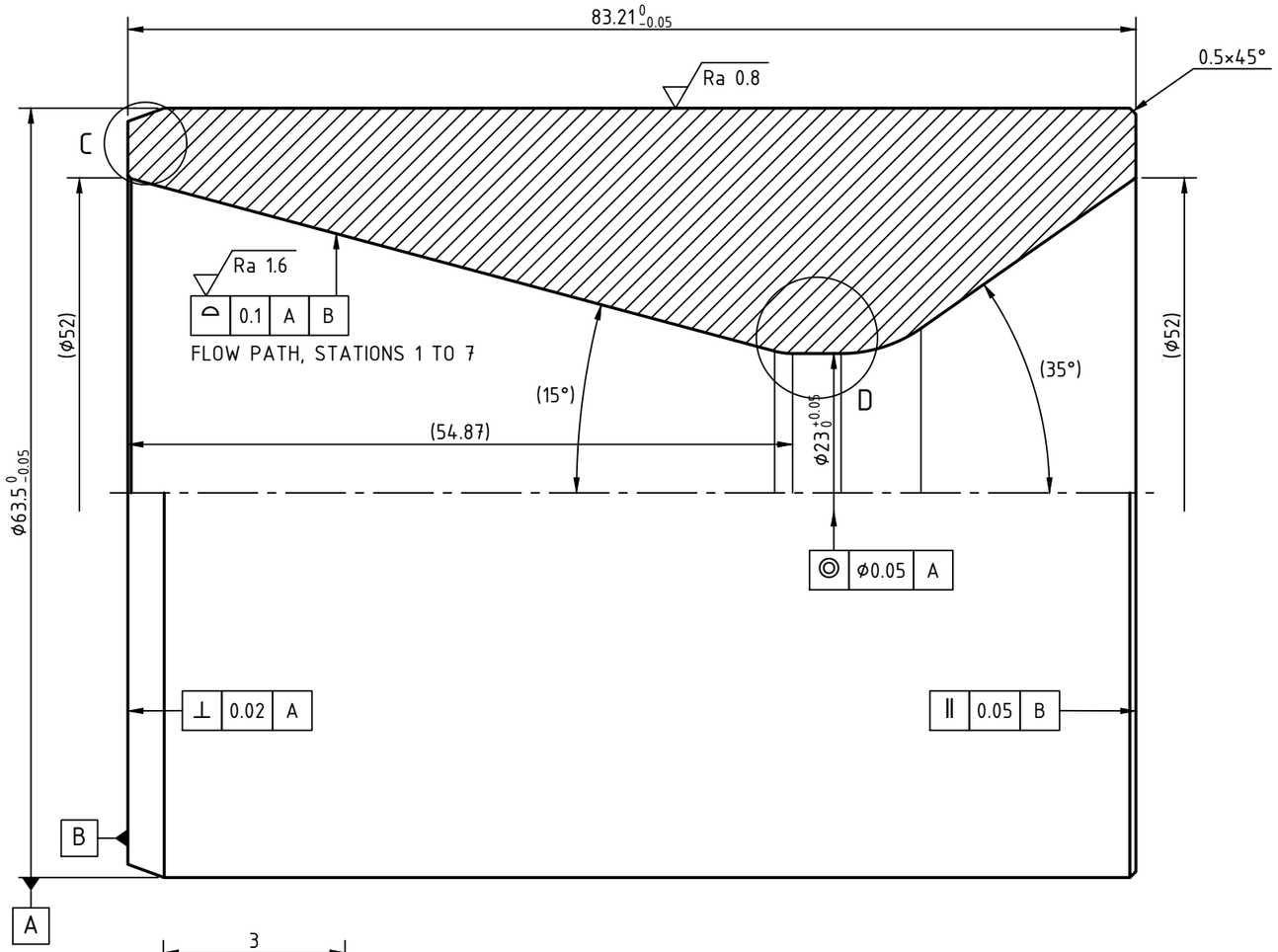
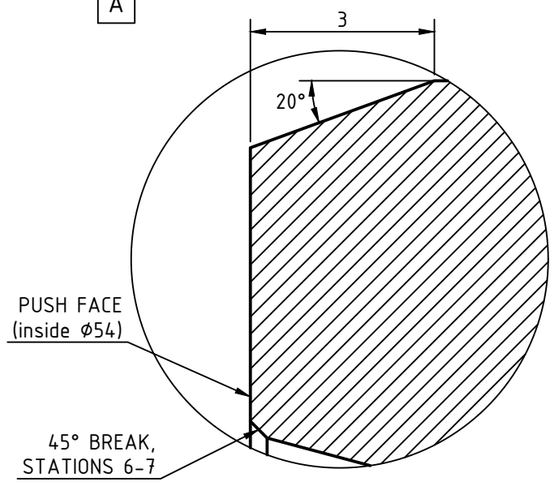
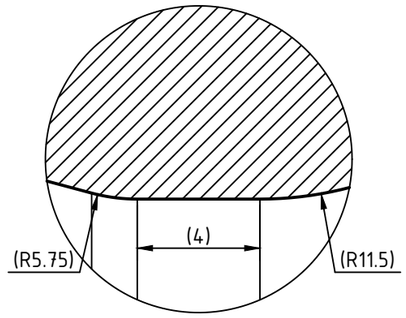




DETAIL C (10:1)



DETAIL D (5:1)

NOTES

1. Material: isostatically moulded fine-grain graphite, grade ISO-63, TTK-4, IG-43 or equal: density ≥ 1.78 g/cm³, flexural strength ≥ 50 MPa, grain ≤ 15 µm; not extruded or vibration-moulded. Certificate required. Blank ϕ68 × 88 min from one block. Mass 311.1 g at 1.80 g/cm³.
2. Machine dry. Never wet this part, never hydro-test it: the proof test uses the steel proof dummy SRM100-120 in its place. No tool marks across the throat land.
3. Edges: forward OD 0.5×45°; aft OD 20° × 3 lead-in; exit 45° break (stations 6-7). Entry edge (station 1) sharp, burr-free, ISO 13715 -0.05, to meet insulator SRM100-105 with no step. Throat blends tangent: do not break. Other edges per the title block.
4. Slip fit in holder SRM100-102 bore ϕ63.58 +0.025/0 (clearance 0.08 to 0.155): by hand, aft end first; never press or knock it. Seated on the lip (seat 83.21 +0.05/0) the forward face lies flush to 0.1 below the holder face. Seal: O-ring AS568-230, FKM 90A (Parker V0709-90 or equal), in the holder gland; grease Parker Super O-Lube or DC-55, thin film.
5. Inspect 100 %: OD and throat at three axial positions, length, contour to the table, coaxiality, perpendicularity, parallelism; 10× visual of the flow path and OD: no cracks, chips or pits. Record against the serial.
6. Laser-mark SRM100-104 and serial on the forward face, between ϕ56 and ϕ62, 2 mm characters, 0.05 deep max (no dot-peen: brittle graphite).
7. Clean with dry, oil-free compressed air only: no liquids, solvents or ultrasonic bath (porous). Handle with clean gloves; store bagged with desiccant.
8. Re-usable. After each firing measure the throat at three axial positions and enter it as THROAT_D (README). Re-use while the throat is ≤ ϕ23.9 and uncracked, with no chips on the flow path or the OD seal surface; otherwise scrap.
9. Used on SRM100-000 item 3. Not fitted on proof assembly SRM100-001 (proof dummy SRM100-120 in its place).

FLOW-PATH CONTOUR (THEORETICALLY EXACT)

STATION	z FROM B	ϕ	NOTE
1	83.206	52.000	entry, forward face: sharp edge (note 3)
2	65.468	27.160	35° cone / R11.5 tangent
3	58.872	23.000	R11.5 / land tangent (throat area 415.5 mm²)
4	54.872	23.000	land / R5.75 tangent
5	53.384	23.392	R5.75 / 15° cone tangent
6	0.273	51.854	15° cone / 45° exit break
7	0.000	52.400	45° exit break / aft face B
E	0.000	52.000	exit, 15° cone meets B (construction; area ratio 5.11)
C1	58.872	46.000	centre of R11.5
C2	54.872	34.500	centre of R5.75

Table values are TEDs (profile frame); values in parentheses in the views are auxiliary.

Ra 3.2 (✓)

MATERIAL: Isomolded graphite, ISO-63 class	MASS: 311.1 g	GENERAL TOLERANCES: ISO 2768-mK	SURFACE TEXTURE: Ra 3.2	EDGES: ISO 13715 -0.3 / +0.3
DOCUMENT TYPE: DETAIL DRAWING	DOCUMENT STATUS: NOT CHECKED - FOR REVIEW	SCALE: 2:1	PROJECTION	UNITS: mm
LEGAL OWNER: HELIOS SRM-100 SOLID ROCKET MOTOR	TITLE, SUPPLEMENTARY TITLE: NOZZLE INSERT — Throat ϕ23 - 35° / 15° conical de Laval - graphite			
CREATED BY: text-to-cad / Claude	APPROVED BY: -	DATE OF ISSUE: 2026-09-23	IDENTIFICATION NUMBER: SRM100-104	REV: A / SHEET: 1 / 1